JOINS OF GROTHENDIECK TOPOLOGIES OF IDEAL TYPE IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: THE STABILISATION CRITERION

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Abstract. For a small category S, every idempotent two-sided ideal D of morphisms defines a Grothendieck topology J_D, covering sieves being those containing $D(-,c)$, and every topology J defines the class W_J of maps of presheaves inverted by sheafification, an element of the complete lattice Same(Presh(S)) of classes of morphisms closed under composition and two-out-of-three. The map $J \mapsto W_J$ preserves all meets; it preserves the join of two topologies on every finite category, but not on the poset of natural numbers. We prove the exact criterion for topologies of ideal type on an arbitrary small category: $W_{J_D \vee J_E} = W_{J_D} \vee W_{J_E}$ if and only if the descending chain of alternating ideal products $DE \supseteq DEDE \supseteq \cdots$ stabilises. When it stabilises at I', the join topology is $J_{I'}$ and the proof of the finite case goes through without finiteness; when it does not, a presheaf obtained by gluing two copies of a representable along a sieve, one for each level of the chain, is inverted by the join sheafification but by no bounded alternation of the two local operators. The join of the two classes is contained in the class of maps that are locally bijective along some power $(DE)^k$, a uniform version of locality for the join topology, and the two agree in the stabilising case. For ideals generated by two classes of objects A, B the criterion reads: the join of the classes of maps bijective at the objects of A, respectively of B, is the class of maps bijective at the objects of $A \cap B$ if and only if every morphism factoring through a sufficiently long alternating zigzag of objects of B and A factors through an object of $A \cap B$; on a locally finite poset, if and only if the strictly alternating chains avoiding $A \cap B$ have bounded length.

1. Introduction

Let S be a small category and Presh(S) its category of presheaves. A Grothendieck topology J on S has a sheafification functor a_J, and the class W_J of maps inverted by a_J, the J-local isomorphisms, is a *sameness*: a class of morphisms containing the identities and closed under composition and two-out-of-three. Samenesses form a complete lattice Same(Presh(S))

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under inclusion, meets being intersections and the join of two classes the least sameness containing both [4]. The assignment $J \mapsto W_J$ preserves all meets [6, Thm. 3.1], and the question studied in [6, 7] is whether it preserves binary joins: whether every map inverted by the sheafification of $J \vee J'$ is obtained from J-local and J'-local isomorphisms by finitely many compositions and cancellations. The answer was known to be yes for every finite category [7] and no for the poset $\mathbb{N}$ [6, Thm. 7.1]. This paper gives the exact criterion for topologies of *ideal type*.

A two-sided ideal D of S is a set of morphisms closed under composition with arbitrary morphisms on either side; it is *idempotent* if every member is a composite of two members. Every idempotent ideal D defines a topology J_D whose covering sieves on c are the sieves containing $D(-,c) = \{a \colon x \to c \text{ in } D\}$ [5, Thm. 5.1]; on a finite category every topology is of this form (Kelly–Lawvere [1], [3]), on an infinite one not necessarily. For ideals D, E write $DE = \{d \circ e : d \in D, e \in E\}$, again an ideal, and $(DE)^k = DE \cdots DE$ (k factors of each); the chain $DE \supseteq (DE)^2 \supseteq \cdots$ decreases, since $DE \subseteq D \cap E$.

Theorem 1.1. *Let S be a small category and D, E idempotent ideals of S. Then*

$$W_{J_D \vee J_E} = W_{J_D} \vee W_{J_E} \quad \textit{in } \mathrm{Same}(\mathrm{Presh}(S))$$

if and only if the chain $(DE)^k$ stabilises, i.e. $(DE)^m = (DE)^{m+1}$ for some m. In that case $I' := (DE)^m$ is an idempotent ideal with $(ED)^{m+1} = I'$, $J_D \vee J_E = J_{I'}$, and $W_{J_D} \vee W_{J_E} = W_{J_{I'}}$.

On a finite category the chain stabilises for every pair, which recovers [7]; on $\mathbb{N}$ with the ideals of the even and of the odd points it is strictly decreasing, which recovers [6, Thm. 7.1]. The criterion is the same as for nuclei on a Heyting algebra and their joins in the sameness lattice of fuzzy sets [9], and as the finite-iteration axiom of the lax logic of the join of two nuclei [8]: in all three settings the join of two reflections is generated at the level of inverted maps exactly when the alternation of the two reflections reaches the join in a uniformly bounded number of steps.

The proof has two halves. For the stabilising case, the argument of [7], which used finiteness through the stabilisation of the chain and through a description of ideals by objects of the Cauchy completion, is rewritten without the second ingredient: the idempotency of I' replaces the identities of generating objects. For the non-stabilising case we introduce, for each k, the class U_k of maps that are *locally bijective along* $(DE)^k$, show that the union $U = \bigcup_k U_k$ is a sameness containing W_{J_D} and W_{J_E}, and exhibit, from a morphism in $(DE)^k \setminus (DE)^{k+1}$, a map that is locally bijective along $(DE)^{k+1}$ but not along $(DE)^k$: the fold map of two copies of a representable glued along the sieve $(DE)^{k+1}(-,c)$. The coproduct of these maps over all k is inverted by the join sheafification but lies in no U_k, hence not in the join of the two classes.

The criterion becomes concrete for ideals generated by objects. For a set A of objects, the ideal D_A of morphisms factoring through A is idempotent, and the class $W_A = W_{J_{D_A}}$ consists of the maps of presheaves that are bijective at every object of A: it is the class inverted by restriction to A, the sheafification for the essential subtopos $\mathrm{Presh}(A) \hookrightarrow \mathrm{Presh}(S)$. Two such classes W_A, W_B have the obvious upper bound $W_{A\cap B}$, and Proposition 6.4 applies Theorem 1.1 to decide when the bound is attained: $W_A \vee W_B = W_{A\cap B}$ iff every morphism factoring through a long enough alternating zigzag of objects of B and A factors through an object of $A \cap B$; on a locally finite poset, iff the strictly alternating chains avoiding $A\cap B$ have bounded length. The question is natural because "isomorphism on a class of objects" is the most elementary sameness on a presheaf category, and the answer shows that the join of two of them is in general not of the same kind, even when the corresponding subtoposes meet in $\mathrm{Presh}(A \cap B)$.

What is known. Samenesses and W_J are from [4]; meets of topologies always go to joins of samenesses and the $\mathbb{N}$ counterexample are from [6]; the finite case, with the alternating separated quotients, the plus tower and the retraction of representing bimodules, is [7]. Ideal topologies, the correspondence with idempotent ideals on finite categories and the local operators $R \mapsto \{a : a \cdot D \subseteq R\}$ are in [1, 3, 5]. Local isomorphisms as locally injective and locally surjective maps are standard [2, III.7].

What is new. Theorem 1.1: the necessary and sufficient condition on an arbitrary small category, which decides the question for every pair of ideal topologies, whereas [7] proves the positive answer under the finiteness hypothesis; in particular the failure direction, with the uniform locality classes U_k (Section 3) and the glued representables as witnesses (Section 4); the object-free form of the stabilising direction (Section 5); and the specialisation to ideals generated by objects, Proposition 6.4, where the join of "bijective on A" and "bijective on B" is compared with "bijective on $A \cap B$".

2. Preliminaries

Presheaves are functors $S^{\mathrm{op}} \to \mathbf{Set}$; for $a\colon x \to c$ and $v \in X(c)$ we write $v \cdot a := X(a)v$. A sieve on c is a set of morphisms into c closed under precomposition, i.e. a subpresheaf of the representable $y(c) = S(-, c)$. For an ideal A the sets $A(-, c)$ are sieves and $A(-, c) \cdot b \subseteq A(-, x)$ for $b\colon x \to c$.

Lemma 2.1. *Let D, E be idempotent ideals.*

(i) *J_D is a topology, $D(-, c)$ is its least covering sieve on c, and $J_D \leq J_E$ iff $D \supseteq E$.*
(ii) *A map $f\colon X \to Y$ is in W_{J_D} iff it is* locally bijective along D*: for all c, if $f(x) = f(x')$ for $x, x' \in X(c)$ then $x \cdot a = x' \cdot a$ for all $a \in D(-, c)$, and for all $y \in Y(c)$ and $a \in D(-, c)$, $y \cdot a \in \mathrm{im} f$.*
(iii) *For every k and c the sieve $(DE)^k(-, c)$ is $(J_D \vee J_E)$-covering.*

(iv) *If* $(DE)^m = (DE)^{m+1} =: I'$ *then* I' *is idempotent,* $I' \subseteq D \cap E$, $DI' = I'D = EI' = I'E = I'$, $(ED)^{m+1} = I'$, *every alternating product of at least* $2(m+1)$ *factors* D, E, *in either order, equals* I', *and* $J_D \vee J_E = J_{I'}$. *(The chain* $(ED)^k$ *need not have stabilised at index* m*: Example 6.2.)*

Proof. (i) is [5, Thm. 5.1]: the maximal sieve contains $D(-,c)$; pullback of a sieve containing $D(-,c)$ along $b\colon x \to c$ contains $D(-,x)$ because D is a left ideal; and if $R \supseteq D(-,c)$ and each a^*R' with $a \in R$ contains $D(-,\operatorname{dom} a)$, then for $d \in D(-,c)$, $d = d_1 d_2$ with $d_i \in D$ by idempotency, $d_1 \in R$ and $d_2 \in d_1^* R'$, so $d \in R'$. (ii) is the description of local isomorphisms as locally injective and locally surjective maps [2, III.7, Cor. 6], with the sieves of the definition replaced by the least covering sieve. (iii) Let $K = J_D \vee J_E$. The sieve $D(-,c)$ is K-covering, and for $d \in D(-,c)$ the pullback $d^*\big((DE)(-,c)\big)$ contains $E(-,\operatorname{dom} d)$; by transitivity $(DE)(-,c)$ is K-covering. Inductively, $(DE)^k(-,c)$ is K-covering: it is covered by $(DE)(-,c)$ relative to the sieves $(DE)^{k-1}(-,x)$. (iv) $I'I' = (DE)^{2m} = I'$; $I' \subseteq DE \subseteq D \cap E$; $DI' = DD\,E(DE)^{m-1} = (DE)^m = I'$ as $DD = D$, and similarly $I'E = I'$; $I'D \supseteq I'I' = I'$ as $I' \subseteq D$, and $I'D \subseteq I'$ as I' is a right ideal, so $I'D = I'$, and likewise $EI' = I'$; then $(ED)^{m+1} = E(DE)^m D = EI'D = I'$. An alternating product of at least $2(m+1)$ factors contains $(DE)^{m+1}$ or $(ED)^{m+1}$ as a block of consecutive factors; the block equals I', and multiplying I' by D or E on either side returns I'. For the join, $J_{I'} \geq J_D, J_E$ since $I' \subseteq D, E$, and any topology above J_D and J_E has $I'(-,c) = (DE)^m(-,c)$ covering by (iii), hence is above $J_{I'}$. □

Theorem 2.2 (join theorem, [5, Thm. 4.1])**.** *Let* U, V *be samenesses on a category* C *whose join contains the isomorphisms, and let* $P\colon C \to C$ *be a functor with a natural transformation* $\pi\colon \mathrm{id} \Rightarrow P$ *such that* $\pi_X \in U \vee V$ *for all* X *and* P *inverts* U *and* V*. Then* $U \vee V = W_P$*, the class of maps inverted by* P*.*

3. Uniform locality along the chain

Fix idempotent ideals D, E, write $K = J_D \vee J_E$ and $V = W_{J_D} \vee W_{J_E}$.

Definition 3.1. For $k \geq 1$ let U_k be the class of maps $f\colon X \to Y$ that are locally bijective along the ideal $(DE)^k$ in the sense of Lemma 2.1(ii): $f(x) = f(x')$ implies $x \cdot a = x' \cdot a$ for all $a \in (DE)^k(-,c)$, and $y \cdot a \in \mathrm{im} f$ for all $y \in Y(c)$ and $a \in (DE)^k(-,c)$. Put $U = \bigcup_k U_k$.

Lemma 3.2. $U_k \subseteq U_{k+1}$, U *is a sameness,* $W_{J_D} \cup W_{J_E} \subseteq U_1$, *and* $U \subseteq W_K$. *Hence* $V \subseteq U \subseteq W_K$.

Proof. $U_k \subseteq U_{k+1}$ because $(DE)^{k+1} \subseteq (DE)^k$. Identities lie in every U_k. Let $h\colon X \to Y$ and $g\colon Y \to Z$.

Composition. If $h \in U_k$ and $g \in U_l$ then $gh \in U_{k+l}$: every $a \in (DE)^{k+l}(-,c)$ is $a = be$ with $b \in (DE)^l$, $e \in (DE)^k$. If $gh(x) = gh(x')$

then $h(x)\cdot b = h(x')\cdot b$, i.e. $h(x\cdot b) = h(x'\cdot b)$, so $x\cdot be = x'\cdot be$. For $z \in Z(c)$, $z\cdot b = g(y)$ for some y, and $y\cdot e = h(x)$ for some x, so $z\cdot a = gh(x)$.

Cancellation of g. If $gh \in U_k$ and $g \in U_k$ then $h \in U_{2k}$. Local injectivity of h along $(DE)^k$ follows from that of gh. For $y \in Y(c)$ and $a = be$ with $b, e \in (DE)^k$: $g(y)\cdot b = gh(x)$ for some x, so $g(y\cdot b) = g(h(x))$ and, g being locally injective along $(DE)^k$, $y\cdot be = h(x)\cdot e \in \mathrm{im}h$.

Cancellation of h. If $gh \in U_k$ and $h \in U_k$ then $g \in U_{2k}$. Local surjectivity of g along $(DE)^k$ follows from that of gh. If $g(y) = g(y')$ and $a = be$ with $b, e \in (DE)^k$, choose x, x' with $h(x) = y\cdot b$ and $h(x') = y'\cdot b$; then $gh(x) = g(y)\cdot b = g(y')\cdot b = gh(x')$, so $x\cdot e = x'\cdot e$ and $y\cdot a = h(x\cdot e) = h(x'\cdot e) = y'\cdot a$.

Hence U is a sameness. $W_{J_D} \subseteq U_1$: a J_D-local isomorphism is locally bijective along $D \supseteq DE$. $W_{J_E} \subseteq U_1$: $DE \subseteq E$ because E is a left ideal. Finally $U_k \subseteq W_K$: a map in U_k is K-locally injective and K-locally surjective because $(DE)^k(-, c)$ is K-covering (Lemma 2.1(iii)). □

Remark 3.3. U is the class of maps that are K-local isomorphisms *uniformly*: the covering sieves witnessing local injectivity and surjectivity can be taken to be $(DE)^k(-, c)$ for one k independent of the element. W_K only requires, for each element, a K-covering sieve, and K-covering sieves are built from the sieves $(DE)^k(-, c)$ by transitivity with unbounded k.

4. Failure without stabilisation

Proposition 4.1. *Suppose $a \in (DE)^k \setminus (DE)^{k+1}$, $a\colon x \to c$. Let X be the presheaf obtained from two copies $y(c)^{(1)}, y(c)^{(2)}$ of the representable by identifying $b^{(1)}$ with $b^{(2)}$ for every $b \in (DE)^{k+1}(-, c)$, and let $f\colon X \to y(c)$ be the fold map. Then $f \in U_{k+1} \setminus U_k$.*

Proof. The identification is compatible with restriction because $(DE)^{k+1}(-, c)$ is a sieve, so X is a presheaf, the pushout of $y(c) \leftarrow (DE)^{k+1}(-, c) \to y(c)$, and f is a map of presheaves, surjective. If $f(u) = f(u')$ with $u \neq u'$ then $u = b^{(1)}$, $u' = b^{(2)}$ for one $b \notin (DE)^{k+1}$, and for $e \in (DE)^{k+1}(-, \mathrm{dom}\, b)$ we have $u\cdot e = (be)^{(1)} = (be)^{(2)} = u'\cdot e$ since $be \in (DE)^{k+1}$. So $f \in U_{k+1}$. On the other hand $\mathrm{id}_c \notin (DE)^{k+1}$, for otherwise $a = \mathrm{id}_c \circ a \in (DE)^{k+1}$; so $\mathrm{id}_c^{(1)} \neq \mathrm{id}_c^{(2)}$ in $X(c)$, both map to id_c, and their restrictions along $a \in (DE)^k(-, c)$ are $a^{(1)} \neq a^{(2)}$. Hence $f \notin U_k$. □

Proof of Theorem 1.1, failure direction. Suppose the chain does not stabilise, so that $(DE)^k \supsetneq (DE)^{k+1}$ for every k (once two consecutive terms agree, all later ones do). For each k choose $a_k \in (DE)^k \setminus (DE)^{k+1}$ and let $f_k\colon X_k \to y(c_k)$ be the map of Proposition 4.1. Let $F = \coprod_k f_k\colon \coprod_k X_k \to \coprod_k y(c_k)$. Then $F \in W_K$: it is surjective, and if $F(u) = F(u')$ then u, u' lie in the same summand X_k and agree on the K-covering sieve $(DE)^{k+1}(-, c)$. But $F \notin U_k$ for every k: a map in U_k has all its coproduct components in U_k, since restrictions, images and identifications of elements stay within a

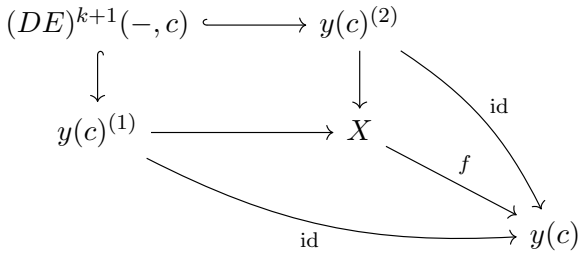


FIGURE 1. The witness of Proposition 4.1: two copies of the representable glued along the sieve $(DE)^{k+1}(-,c)$; the fold map f is locally bijective along $(DE)^{k+1}$ but the two identities, which restrict along $a \in (DE)^k \setminus (DE)^{k+1}$ to distinct elements, are identified by f.

summand, and the component f_k is not in U_k. So $F \in W_K \setminus U \subseteq W_K \setminus V$ by Lemma 3.2, and $V \neq W_K$. □

5. The stabilising case

Assume now $(DE)^m = (DE)^{m+1} = I'$ and put $r := m+1$, so that by Lemma 2.1(iv) I' is idempotent, $(DE)^r = (ED)^r = I'$, every product of at least $2r$ alternating factors D, E in either order equals I', and $K = J_{I'}$. (The shift from m to r is needed: $(ED)^m$ may be strictly larger than I', Example 6.2.) We show $V = W_{J_{I'}}$ by the route of [7], replacing the generating objects used there by the idempotency of I'.

Separated quotients. For an ideal A and $x, x' \in X(c)$ write $x \approx_A x'$ if $x \cdot a = x' \cdot a$ for all $a \in A(-,c)$; this is a congruence of presheaves since A is a left ideal, and $Q_A X := X/\!\approx_A$. The quotient map $q\colon X \to Q_A X$ is locally bijective along A: it is surjective, and $q(x) = q(x')$ means exactly that x, x' agree on $A(-,c)$. For $A = D$ idempotent, $q \in W_{J_D}$.

Lemma 5.1. *Let $D_1 = D$, $D_2 = E$, $D_3 = D, \dots$, $X_0 = X$ and $X_{k+1} = Q_{D_{k+1}} X_k$. Two elements of $X(c)$ have the same image in X_k iff they agree on $(D_k \cdots D_1)(-,c)$; hence $X_{2r} = Q_{I'} X$, the composite $X \twoheadrightarrow Q_{I'} X$ lies in V, and $Q_{I'} X$ is $J_{I'}$-separated.*

Proof. Induction on k: x, x' have the same image in X_{k+1} iff $x \cdot b$ and $x' \cdot b$ have the same image in X_k for all $b \in D_{k+1}(-,c)$, i.e. iff $x \cdot ba = x' \cdot ba$ for all such b and all $a \in (D_k \cdots D_1)(-, \operatorname{dom} b)$, and these ba are the elements of $(D_{k+1} D_k \cdots D_1)(-,c)$. For $k = 2r$ this ideal is $(ED)^r = I'$. Each quotient map lies in $W_{J_{D_{k+1}}} \subseteq V$. If two elements of $Q_{I'} X(c)$ agree on $I'(-,c)$, their representatives x, x' satisfy $x \cdot a \approx_{I'} x' \cdot a$ for $a \in I'(-,c)$, i.e. $x \cdot ab = x' \cdot ab$ for $a, b \in I'$; as $I' = I'I'$ this says $x \approx_{I'} x'$. □

The plus tower. An S–S-bimodule is a functor $S^{\mathrm{op}} \times S \to \mathbf{Set}$; ideals are sub-bimodules of Hom_S, and bimodules have a tensor product $(A \otimes B)(x,z) = \coprod_y A(y,z) \times B(x,y)/\sim$, with $\mathrm{Hom}(A \otimes B, X) \cong \mathrm{Hom}(A, \mathrm{Hom}(B,X))$ for the presheaf $\mathrm{Hom}(A,X)(c) = \mathrm{Hom}_{\mathrm{Presh}(S)}(A(-,c),X)$. Since $D(-,c)$ is the least J_D-covering sieve, the plus construction for J_D is $X^{+D} = \mathrm{Hom}(D,X)$ with unit $x \mapsto (d \mapsto x \cdot d)$, and the alternating plus tower is $\Phi_k X = \mathrm{Hom}(T_k, X)$, $T_k = D_k \otimes \cdots \otimes D_1$, with unit $u_k \colon X \to \Phi_k X$ a composite of maps of $W_{J_D} \cup W_{J_E}$; and $a_{I'}X = X^{+I'+I'} = \mathrm{Hom}(P,X)$, $P = I' \otimes I'$, with unit η. None of this uses objects of S. The two bimodule lemmas of [7, Lemmas 2.2, 2.3] are reproved here in the present generality.

Lemma 5.2. (a) *The map $m_k \colon P \otimes T_k \to P$, $[i, i', d_k, \ldots, d_1] \mapsto [i,\, i'd_k \cdots d_1]$, is a well-defined map of bimodules, $m_k^* = a_{I'}(u_k)$ under $a_{I'}\Phi_k X = \mathrm{Hom}(P, \mathrm{Hom}(T_k, X)) = \mathrm{Hom}(P \otimes T_k, X)$, and m_k is an isomorphism.*

(b) *Let $k \geq 4r$ and $2r \leq j \leq k-2r$. The map $r_k \colon T_k \to P$, $[d_k, \ldots, d_1] \mapsto [d_k \cdots d_{j+1},\ d_j \cdots d_1]$, is a well-defined map of bimodules, and $r_k \circ \varepsilon_k = m_k$, where $\varepsilon_k \colon P \otimes T_k \to T_k$, $[p,t] \mapsto \bar{p} \cdot t$ ($\bar{p} = ii'$ for $p = [i,i']$), is the bimodule map inducing the sheafification unit $\Phi_k X \to a_{I'}\Phi_k X$.*

Consequently the comparison $\iota_k := (m_k^*)^{-1} \circ \varepsilon_k^* \colon \Phi_k X \to a_{I'}X$, the sheafification unit of $\Phi_k X$ followed by $(a_{I'}u_k)^{-1}$, satisfies $\iota_k \circ r_k^* = \mathrm{id}$, so is surjective for $k \geq 4r$, and $\iota_k \circ u_k = \eta_X$.

Proof. (a) The value $i'd_k \cdots d_1$ lies in I' because I' is a right ideal; a tensor move inside $[d_k, \ldots, d_1]$ or between i and i' does not change the image, and a move $[i, i'n, d_k, \ldots] \sim [i, i', nd_k, \ldots]$ gives $[i, i'nd_k \cdots d_1]$ either way. Compatibility with the actions of S is clear. For $\psi \in \mathrm{Hom}(P(-,c), X)$, the element $a_{I'}(u_k)(\psi) = u_k \circ \psi$ sends p to $(t \mapsto \psi(p) \cdot \bar{t}) = (t \mapsto \psi(p \cdot \bar{t}))$, and $p \cdot \bar{t} = m_k[p,t]$; so $a_{I'}(u_k) = m_k^*$. Since $u_k \in V \subseteq W_{J_{I'}}$, $a_{I'}(u_k)$ is bijective for every X and c, hence $m_k(-,c)$ is an isomorphism of presheaves by the Yoneda lemma, for every c. (b) The two blocks $d_k \cdots d_{j+1}$ and $d_j \cdots d_1$ are alternating products of $k - j \geq 2r$ and $j \geq 2r$ factors, hence lie in I' by Lemma 2.1(iv). A tensor move inside a block does not change its product, and a move across the boundary, $[\ldots, d_{j+1}n, d_j, \ldots] \sim [\ldots, d_{j+1}, nd_j, \ldots]$, changes the pair of products from (an, b) to (a, nb) with $a, b \in I'$, which are identified in $P = I' \otimes I'$. So r_k is well defined, and it is a bimodule map. The unit $y \mapsto (p \mapsto y \cdot \bar{p})$ of $a_{I'}$ at $Y = \Phi_k X$ sends φ to $(p \mapsto (t \mapsto \varphi(\bar{p} \cdot t))) = \varepsilon_k^*(\varphi)$. For $p = [i,i']$ and $t = [d_k, \ldots, d_1]$,

$$r_k \varepsilon_k [p,t] = [ii'd_k \cdots d_{j+1},\ d_j \cdots d_1] = [i,\, i'd_k \cdots d_1] = m_k[p,t],$$

the middle equality being the tensor move of $i'd_k \cdots d_{j+1}$ across the tensor sign of P, legitimate as i and $d_j \cdots d_1$ lie in I'. Finally $\iota_k r_k^* = (m_k^*)^{-1}(r_k\varepsilon_k)^* = (m_k^*)^{-1} m_k^* = \mathrm{id}$, and $\iota_k u_k = \eta_X$ is the naturality square $\eta_{\Phi_k X} \circ u_k = a_{I'}(u_k) \circ \eta_X$ of the unit η. □

Lemma 5.3. *Let Y be $J_{I'}$-separated. Then Y^{+D} and Y^{+E} are $J_{I'}$-separated. Hence every $\Phi_k Y$ is $J_{I'}$-separated, ι_k is injective, bijective for $k \geq 4r$, and $\eta_Y \in V$.*

Proof. Let $\varphi, \psi \in Y^{+D}(c) = \mathrm{Hom}(D(-,c), Y)$ agree on $I'(-,c)$, i.e. $\varphi \cdot p = \psi \cdot p$ for all $p \in I'(-,c)$, where $(\varphi \cdot p)(n) = \varphi(pn)$ for $n \in D(-, \mathrm{dom}\, p)$. Let $n \in D(-,c)$, $n\colon x \to c$, and $b \in I'(-,x)$. Then $nb \in DI' = I' = I'I'$, so $nb = pq$ with $p, q \in I'$ and $q \in D(-, \mathrm{dom}\, p)$, and

$$\varphi(n) \cdot b = \varphi(nb) = \varphi(pq) = (\varphi \cdot p)(q) = (\psi \cdot p)(q) = \psi(nb) = \psi(n) \cdot b.$$

As Y is $J_{I'}$-separated, $\varphi(n) = \psi(n)$; so $\varphi = \psi$. The same holds for E. Hence all $\Phi_k Y$ are $J_{I'}$-separated, and ι_k, the sheafification unit of a separated presheaf followed by an isomorphism, is injective [2, III.5]; it is surjective for $k \geq 4r$ by the section r_k^* of Lemma 5.2. So ι_k is an isomorphism, and $\eta_Y = \iota_k u_k \in V$, as isomorphisms lie in W_{J_D}. □

Proof of Theorem 1.1, stabilising direction. Apply Theorem 2.2 with $U = W_{J_D}$, $V' = W_{J_E}$, $P = a_{I'}$ and $\pi = \eta$; $a_{I'}$ inverts both classes since $J_D, J_E \leq J_{I'}$. For a presheaf X the unit factors as $X \twoheadrightarrow Q_{I'}X \xrightarrow{\eta} a_{I'}Q_{I'}X \cong a_{I'}X$, the first map in V by Lemma 5.1, the second by Lemma 5.3, and the isomorphism induced by the quotient map, which $a_{I'}$ inverts. So $\eta_X \in V$ for all X, and $V = W_{a_{I'}} = W_{J_{I'}} = W_K$. □

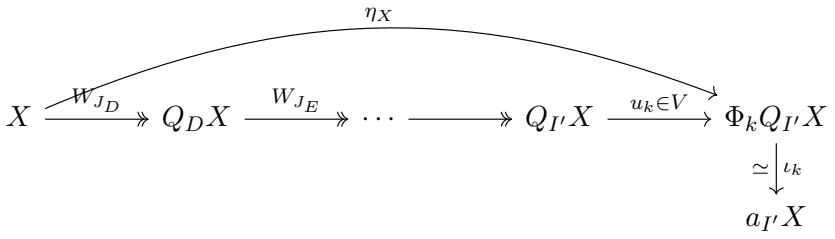


FIGURE 2. The stabilising direction: $2r$ separated quotients and $4r$ plus constructions connect a presheaf to its join sheafification inside $V = W_{J_D} \vee W_{J_E}$.

6. Examples and remarks

Example 6.1. On a finite category every chain of ideals stabilises, so Theorem 1.1 contains [7, Thm. 1.1], with the bound $2r$ quotients and $4r$ plus constructions, $r = m+1$ (in [7] the index is chosen so that both chains $(DE)^k$ and $(ED)^k$ have stabilised, which is the same bound). On $S = (\mathbb{N}, \leq)$ with D and E the ideals of arrows $n \to m$ passing through an even, respectively an odd, point, $(DE)^k$ consists of the arrows $n \to m$ admitting k alternating even and odd points between n and m, a strictly decreasing chain with empty intersection; the join topology is degenerate and W_K is everything,

while V is not [6, Thm. 7.1]. The witnesses of Proposition 4.1 are here the fold maps of two copies of $y(c)$ glued below $c-2k-1$, and their coproduct is a variant of the presheaf $n \mapsto \{x \geq n\}$ used there.

Example 6.2. The two chains $(DE)^k$ and $(ED)^k$ need not stabilise at the same index. Let S be the category with two objects $0, 1$ and one non-identity arrow $a\colon 0 \to 1$, and let $D = \{\mathrm{id}_0, a\}$, $E = \{\mathrm{id}_1, a\}$ be the ideals of arrows factoring through 0, respectively through 1; both are idempotent. Then $DE = \emptyset$ (no arrow of E has codomain 0), while $ED = \{a\}$. So $(DE)^1 = (DE)^2 = \emptyset = I'$, $m = 1$, but $(ED)^1 = \{a\} \neq I'$, and only $(ED)^2 = \emptyset = I'$. Accordingly, for the constant presheaf X with value a two-element set, the second separated quotient $Q_E Q_D X = Q_{ED} X$ still has two elements at 1, whereas $Q_{I'} X = Q_\emptyset X$ is terminal: the bound $2m = 2$ of quotients is insufficient, and Lemma 5.1 uses the bound $2r = 4$ (here three quotients already suffice, as $DED = \emptyset$; $2r$ is a bound, not the minimum). The join $J_D \vee J_E = J_\emptyset$ is the degenerate topology, as the empty sieve on 0 is J_E-covering and the sieve $\{a\}$ on 1 is J_D-covering.

Example 6.3. For a monoid M, viewed as a category with one object, ideals are the two-sided ideals of M and the criterion is the stabilisation of the alternating products $DE \supseteq DEDE \supseteq \cdots$ in the quantale of ideals of M. Every finite monoid satisfies it, with m at most the number of elements; [7, §7] gives monoids of order 8 and 12 on which the plus tower alone does not stabilise although the chain does. For infinite monoids the chain can decrease forever, and then Theorem 1.1 says that the join of the two classes of local isomorphisms of M-sets is strictly smaller than the class for the join topology.

Ideals generated by objects. For a set A of objects of S let D_A be the set of morphisms factoring through an object of A. It is an ideal, idempotent since $f = g \circ h$ with $h\colon x \to a$, $g\colon a \to y$ has $g, h \in D_A$; on a finite category every idempotent ideal is of this form after Cauchy completion [7, §2], but not in general: on $(\mathbb{Q}, \leq)$ the ideal of strict inequalities is idempotent and contains no identity. Write $W_A := W_{J_{D_A}}$.

Proposition 6.4. *Let A, B be sets of objects of S.*

(a) *W_A is the class of maps f with f_a bijective for every $a \in A$; $(D_A D_B)^k$ is the set of morphisms factoring through a chain $b_1 \to a_1 \to b_2 \to \cdots \to b_k \to a_k$ with $b_i \in B$, $a_i \in A$, and contains $D_{A \cap B}$.*

(b) *$W_A \vee W_B = W_{A \cap B}$ if and only if for some N every morphism factoring through such a chain of length N factors through an object of $A \cap B$, i.e. $(D_A D_B)^N \subseteq D_{A \cap B}$.*

(c) *If $S = P$ is a poset whose intervals are finite, the condition in (b) holds if and only if the strictly alternating chains $b_1 < a_1 < \cdots < b_k < a_k$ ($b_i \in B$, $a_i \in A$) contained in intervals disjoint from $A \cap B$ have a bounded number of elements; if they do not, the chain $(D_A D_B)^k$ does not stabilise and $W_A \vee W_B \subsetneq W_{J_{D_A} \vee J_{D_B}}$.*

Proof. (a) If f_a is bijective for $a \in A$, then for $x, x' \in X(c)$ with $f(x) = f(x')$ and $g = g_1 g_2 \in D_A(-, c)$ with $g_1 \colon a \to c$, $a \in A$, we get $f_a(x \cdot g_1) = f_a(x' \cdot g_1)$, so $x \cdot g_1 = x' \cdot g_1$ and $x \cdot g = x' \cdot g$; and for $y \in Y(c)$, $y \cdot g_1 = f_a(z)$ for some z, so $y \cdot g = f(z \cdot g_2)$. Thus f is locally bijective along D_A, and $f \in W_A$ by Lemma 2.1(ii). Conversely $\mathrm{id}_a \in D_A(-, a)$ for $a \in A$, so local bijectivity along D_A gives f_a bijective. A morphism of $D_A D_B$ is $d \circ e$ with e factoring through some $b \in B$ and d through some $a \in A$, i.e. a morphism factoring through a chain $b \to a$; iterating gives the description of $(D_A D_B)^k$, and a morphism through $c \in A \cap B$ factors through the chain $c \to c \to \cdots \to c$ of identities. (b) If $(D_A D_B)^N \subseteq D_{A \cap B}$ then $(D_A D_B)^N \subseteq D_{A \cap B} \subseteq (D_A D_B)^{N+1} \subseteq (D_A D_B)^N$, so the chain stabilises at $I' = D_{A \cap B}$ and Theorem 1.1 gives $W_A \vee W_B = W_{J_{I'}} = W_{A \cap B}$. Conversely, $D_{A \cap B} \subseteq D_A \cap D_B$ gives $J_{D_{A \cap B}} \geq K := J_{D_A} \vee J_{D_B}$ by Lemma 2.1(i), hence $W_{A \cap B} \supseteq W_K \supseteq W_A \vee W_B$. If $W_A \vee W_B = W_{A \cap B}$, then $W_A \vee W_B = W_K$, the chain stabilises at some $I' = (D_A D_B)^m$ by Theorem 1.1, and $W_{J_{I'}} = W_K = W_{J_{D_{A \cap B}}}$. A topology is determined by the class of maps inverted by its sheafification, since the sheaves are the objects local with respect to that class (a reflective subcategory is recovered from the maps its reflector inverts) and subtoposes of $\mathrm{Presh}(S)$ correspond bijectively to topologies [2, Chap. V]; so $J_{I'} = J_{D_{A \cap B}}$, $I' = D_{A \cap B}$ by Lemma 2.1(i), and $N = m$ works. (c) In a poset, $(n \leq m) \in (D_A D_B)^k$ iff there are $n \leq b_1 \leq a_1 \leq \cdots \leq b_k \leq a_k \leq m$ with $b_i \in B$, $a_i \in A$; if the interval $[n, m]$ is disjoint from $A \cap B$, an equality $b_i = a_i$ or $a_i = b_{i+1}$ would produce a point of $A \cap B$, so the chain is strict. Hence $(D_A D_B)^k \setminus D_{A \cap B}$ consists of the pairs $n \leq m$ with $[n, m]$ disjoint from $A \cap B$ and containing a strictly alternating chain with $2k$ elements. If these chains have at most L elements, then $(D_A D_B)^k = D_{A \cap B}$ for $2k > L$. If their sizes are unbounded, $(D_A D_B)^k \neq D_{A \cap B}$ for every k, while $\bigcap_k (D_A D_B)^k = D_{A \cap B}$ because a chain with more elements than the finite interval $[n, m]$ is not strict; a descending chain whose members all differ from its intersection does not stabilise, and Theorem 1.1 gives the strict inclusion. □

Example 6.5. On $P = (\mathbb{N}, \leq)$ let A be the even numbers and B the odd numbers together with the multiples of 4, so $A \cap B = 4\mathbb{N}$. An interval disjoint from $4\mathbb{N}$ lies in some $[4j+1, 4j+3]$, whose longest strictly alternating chain is $4j+1 < 4j+2 < 4j+3$; so $L = 3$, $(D_A D_B)^2 = D_{4\mathbb{N}}$, and a map of presheaves on $\mathbb{N}$ lies in the join of "bijective at the even points" and "bijective at the odd points and at the multiples of 4" if and only if it is bijective at the multiples of 4. With B the odd numbers alone, $A \cap B = \emptyset$, every interval is admissible, the chains are unbounded, and the join is strictly smaller than W_K, which is the class of all maps [6, Thm. 7.1].

Remark 6.6. For a set A of objects, the restriction functor along the full inclusion $i \colon A \hookrightarrow S$ preserves limits and has a fully faithful right adjoint, the right Kan extension Ran_i (and a left adjoint, the left Kan extension). Its essential image is a reflective subcategory whose reflector inverts exactly W_A,

hence it is the subcategory of J_{D_A}-sheaves: $\mathrm{Sh}(S, J_{D_A}) \simeq \mathrm{Presh}(A)$, an essential subtopos of $\mathrm{Presh}(S)$ in the sense of [1]. The meet of the subtoposes $\mathrm{Presh}(A)$ and $\mathrm{Presh}(B)$ is the subtopos of $(J_{D_A} \vee J_{D_B})$-sheaves. Proposition 6.4(b) says that $W_A \vee W_B = W_{A\cap B}$ iff this meet is equivalent to $\mathrm{Presh}(A \cap B)$ *and* is reached by a bounded alternation; the first condition alone does not suffice: for the even and the odd numbers on $\mathbb{N}$ the meet is equivalent to the degenerate subtopos $\mathrm{Presh}(\emptyset)$, yet the join of the samenesses is not $W_\emptyset$.

Remark 6.7. Lemma 3.2 and Theorem 1.1 give three classes $V \subseteq U \subseteq W_K$ with $V = W_K$ iff the chain stabilises, and then $U = V$. Whether $V = U$ holds in general, so that the join of the two classes is exactly the uniformly K-local maps, is open; the reduction lemma of [6, Lemma 4.1], which places every map inverted by $a_{J'}a_J$ in V, is evidence in favour.

Question 6.8. Is $W_{J_D} \vee W_{J_E} = U$ for all idempotent ideals D, E on every small category? Does the criterion extend to topologies that are not of ideal type, with the alternation of the two local operators on Ω in place of the ideal chain?

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